

Blink Rate and Its Relationship to Detecting PTSD, Major Depressive Disorder, and Anxiety Disorders

Executive summary

The strongest direct evidence in this literature is for **depression-spectrum conditions**, not for PTSD or anxiety disorders. Classic observational work found higher blink rates in depressed patients that normalized with clinical improvement, and newer high-resolution eye-tracking work in **late-life depression** again found elevated blink rates relative to healthy older and younger controls, with positive correlations between blink rate and depression/somatic symptom severity. However, a study of **non-retarded, drug-naïve major depression** found **no baseline blink-rate difference** versus controls; instead, blink rate increased after sleep deprivation only in patients, and that increase tracked short-term antidepressant response. Taken together, depression findings suggest that blink rate may reflect a mix of arousal, psychomotor state, autonomic tone, task demands, and possibly treatment-responsive state changes, rather than a single stable trait marker of MDD. ¹

For **PTSD**, the direct blink-rate literature is very thin. The best-targeted study found that women with PTSD showed **higher spontaneous eye-blink rates only under low-threat conditions** during an inhibitory-control task, not a universal elevation across all contexts. For **anxiety disorders**, the clearest direct case-control evidence is a **small panic-disorder pilot** showing higher blink rates than controls at rest and during audiovisual stimulation. I did not find a robust, replicated blink-rate case-control literature for generalized anxiety disorder or social anxiety disorder. ²

Engineering studies show that blink-derived features can contribute to automated depression classification, with reported internal accuracies around **70–75%** for broader eye-movement features and **88–92%** for blink-feature models on structured interview-style tasks. But these are not clinical validation studies: the accessible reports do **not** provide sensitivity/specificity, clinically usable thresholds, or convincing external validation. The current evidence therefore supports blink rate as a **context-sensitive auxiliary feature** in multimodal screening systems, not as a standalone diagnostic biomarker for PTSD, MDD, or anxiety disorders. ³

Methods

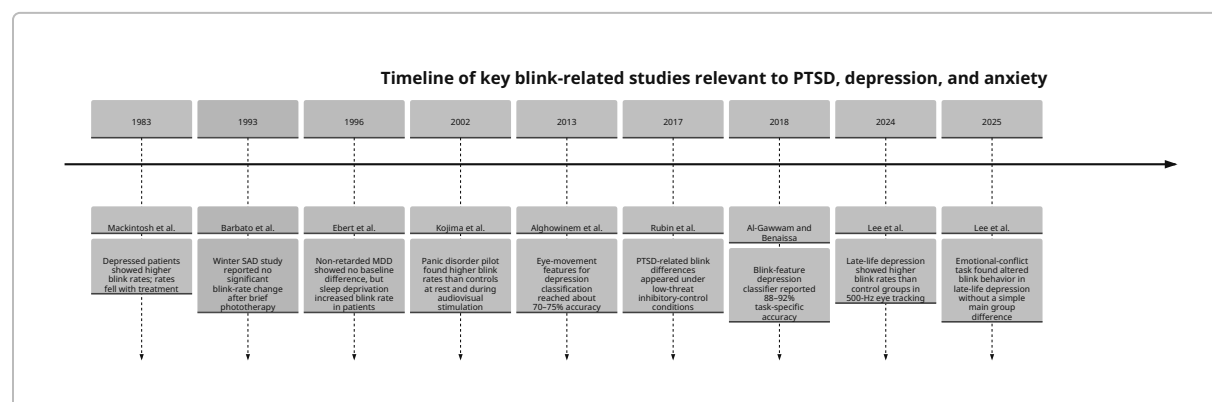
This review covered English-language human studies available up to **May 15, 2026**. I searched for peer-reviewed and high-quality sources indexed or surfaced through **PubMed**, publisher pages, **IEEE**, **ACM**, and related scholarly search results, and I also checked for relevant preprints and technical reports where the peer-reviewed clinical literature was sparse. Search terms combined *blink rate*, *spontaneous eye blink*, *eye blink response*, *PTSD*, *major depression*, *late-life depression*, *panic disorder*, *anxiety*, *eye tracking*, *EOG*, *classifier*, and *psychophysiology*. I excluded animal-only studies, non-English papers, and studies focused only on **startle-blink magnitude/EMG** without a spontaneous or task-related blink-rate measure.

I prioritized sources that reported at least some combination of: diagnosis/population, sample size, blink measurement method, task/rest condition, and quantitative or directional findings. Because some older papers were paywalled, several historical studies could only be extracted from structured abstracts, publisher summaries, or journal metadata; those gaps are called out explicitly in the tables as **NR**. I also reviewed the user-provided GitHub note, which documents an **OpenCV/MediaPipe Face Mesh + eye aspect ratio** blink pipeline intended for approximately one-minute clips, with within-clip z-scoring, prominence-based trough detection, and practical advice about lighting, eyeglasses, occlusion, and camera distance.

A key methodological decision was to treat **blink rate** separately from broader ocular phenotypes such as pupil light reflex, emotional gaze bias, microsaccades, and acoustic startle. Those neighboring literatures matter for interpretation and implementation, but they are not equivalent to spontaneous blink rate itself.

Results

The evidence base is small, heterogeneous, and uneven across disorders. Depression has the most direct studies. PTSD has one especially relevant task-based study. Anxiety disorders, outside panic disorder, are strikingly under-studied in the blink-rate literature. Across all three areas, **standardized effect sizes are rarely reported**, and **no retrieved clinical paper provided a validated stand-alone blink-rate threshold with sensitivity/specificity for diagnosis**. 4



The timeline highlights how the field progressed from small observational psychiatry studies to task-based psychophysiology and, more recently, computer-vision classification. 5

PTSD

The most directly relevant PTSD paper is Rubin, Hien, Das, and Melara's 2017 study, **"Inhibitory Control under Threat: The Role of Spontaneous Eye Blinks in Post-Traumatic Stress Disorder."** It included **68 women** divided into PTSD, trauma-exposed without PTSD, and healthy control groups. Blink rate was indexed during a **flanker inhibitory-control task** under varying threat conditions by direct experimenter recording with a digital counter. The accessible full-text lines confirm the three-group design and total sample, and the abstract summary reports the core result: **women with PTSD maintained higher blink rates than controls under low threat**, whereas this was not a stable across-the-board disorder effect under all conditions. The authors interpreted the finding as consistent with poorer default working-

memory/inhibitory-control state under low-threat conditions, rather than a simple universal PTSD blink-rate signature. No diagnostic threshold, sensitivity, or specificity was reported. ⁶

This is important for feasibility. It suggests that blink rate in PTSD may be **state- and context-dependent**, meaning that a passive resting measure might miss the signal entirely. It also means that task design matters: the blink difference emerged in a threat-manipulated cognitive-control paradigm, not as a general-purpose clinical threshold. That sharply limits transportability to screening settings unless the eliciting context is standardized. ⁷

Major depressive disorder

Depression is where the blink-rate literature is both richer and more contradictory. The classic 1983 British Journal of Psychiatry paper by Mackintosh, Kumar, and Kitamura studied **23 depressed patients** and a **matched normal group**, quantified behavior from **video recordings** across **three interviews**, and concluded that **blink rate was increased in depression and fell to normal levels during treatment**. The paper explicitly states that the blink-rate effect was **independent of medication** but related to clinical improvement. This is one of the clearest historical signals in favor of a depression-related blink abnormality. ⁸

But the 1996 Neuropsychopharmacology study by Ebert and colleagues complicates that story. In **12 drug-naïve patients with non-retarded major depression** and **12 healthy controls**, blink rates were compared before and after **sleep deprivation** and before and after a **2.5 mg bromocriptine challenge**. Here, **baseline eye-blink rate was not different** between patients and controls. Instead, **sleep deprivation increased blink rate in depressed patients but not in controls**, and the size of the increase was proportional to improvement in depressive state. Bromocriptine did not increase blink rate one hour after administration. This makes blink rate look less like a universal MDD trait marker and more like a state-sensitive index tied to psychomotor subtype, arousal, or short-term treatment response. ⁹

Historical depressive-subtype evidence is also mixed. Barbato and colleagues' 1993 paper on **winter seasonal affective disorder** reported that **brief morning phototherapy for one week did not significantly change mean blink rate in 10 SAD patients**; the accessible summary gives the patient means as approximately **13/min before** versus **10/min after**, without statistical significance. This is a useful negative result: even within a depressive subtype with a plausible dopaminergic/light-regulation hypothesis, blink rate did not behave like a clean state marker. ¹⁰

The most methodologically modern depression evidence comes from **late-life depression (LLD)**. Lee and colleagues' 2024 BMC Geriatrics study included **25 LLD patients**, **29 age-matched healthy older adults**, and **25 healthy younger adults**. Eye position, pupil size, and blink rate were measured with an **EyeLink 1000 Plus** at **500 Hz** in a dark room while researchers manipulated **background luminance** to elicit pupil light and darkness reflexes. LLD patients had **significantly higher blink rates** than both healthy groups, and blink rate in older participants correlated positively with **GDS-15** and **PHQ-15** scores. Medications were **not discontinued**, but the authors used mixed models to test medication effects and reported that group differences or trends in blink rate persisted after accounting for medication type. The study still did not report a clinically actionable blink threshold or diagnostic sensitivity/specificity. ¹¹

A 2025 follow-up from the same research line, **"Pupil and Eye Blink Response Abnormalities During Emotional Conflict Processing in Late-Life Depression,"** reported that LLD patients showed significantly

different pupil and blink behavior during an **emotional conflict** task. The accessible abstract summary is crucially more nuanced than the 2024 luminance study: blink rates were **higher in incongruent than congruent conditions**, but the available summary indicates **no simple significant group difference in blink rates**. In other words, once the task changed from reflexive luminance modulation to emotional conflict processing, blink became more of a **task-modulated process marker** than a straightforward group discriminator. ¹²

Two engineering studies are relevant because they move from association to detection. Alghowinem and colleagues' 2013 IEEE paper on **eye movement analysis for depression detection** reported **about 70% accuracy** using low-level eye-movement features and **about 75% accuracy** using statistical measures with SVM-based classification in a binary depressed/non-depressed setting. Accessible snippets also report that **blink duration** features were longer in depressed subjects. Al-Gawwam and Benaissa's 2018 IEEE blink-feature paper reported that **AdaBoost** outperformed other tested classifiers and achieved roughly **92% accuracy** on the **NorthWind** task and **88% accuracy** on the **FreeForm** task. These are interesting proof-of-concept results, but they do not amount to clinical validation: the accessible reports do not expose sensitivity/specificity, and these task-specific accuracies are still vulnerable to overfitting to structured interview conditions. ³

A historical treatment paper by Berrios and Canagasabay also belongs in the record. The accessible abstract confirms that they compared the effects of **ECT** and **antidepressant drugs** on blink rate and psychomotor retardation in **two matched groups of depressed patients**, but the quantitative blink findings were not exposed in the accessible abstract summary I could retrieve, so I do not rely on that paper for the main synthesis. ¹³

Anxiety disorders

Direct blink-rate evidence for anxiety disorders is thin, and most of it concentrates in **panic disorder**. Kojima and colleagues' 2002 Psychiatry and Clinical Neurosciences paper examined blink rate in **11 male patients with panic disorder** and **16 age-matched normal controls** under **rest** and **audiovisual stimulation** involving videos of imagined experiences such as driving or diving into the sea. The study found that **blink rate was significantly higher in panic-disorder patients under both conditions**, with a particularly notable elevation during the **sea scene** intended as a relaxation condition. The authors concluded that blink rate *may* have potential for assessing anxiety state, but they were explicit that the study was **preliminary** and based on a **small sample**. No threshold or sensitivity/specificity was reported. ¹⁴

What matters most is what I **did not** find: I did not locate a comparably robust, replicated body of blink-rate case-control studies for generalized anxiety disorder, social anxiety disorder, or other anxiety-spectrum diagnoses. That absence is itself informative. At present, blink rate cannot be said to have the same level of disorder-specific empirical support in anxiety disorders that it has in depression. The panic-disorder pilot suggests a signal is possible, but not yet stable or generalizable enough for detection claims. ¹⁴

Source summary table

The table below summarizes the core disorder-relevant sources and the most usable methodological details.

Source	DOI	Study type	Sample and population	Blink measurement method	Condition
Mackintosh, Kumar, & Kitamura, 1983, <i>Br J Psychiatry</i>	10.1192/BJP.143.1.55	Longitudinal case-control	23 depressed patients + matched normal group	Behavior quantified from video recordings	Three standard intervals during treatment
Ebert et al., 1996, <i>Neuropsychopharmacology</i>	10.1016/0893-133X(95)00237-8	Drug-naive case-control + challenge study	12 non-retarded MDD patients + 12 healthy controls	Blink-rate measurement method not exposed in abstract; repeated measures before/after interventions	Baseline sleep deprivation after 2 bromide
Berrios & Canagasabay, 1990, <i>Convulsive Therapy / Journal of ECT</i>	DOI not exposed in accessible summary	Matched treatment comparison	Two matched groups of depressed patients	NR in accessible abstract	ECT vs. antidepressant treatment comparison

Source	DOI	Study type	Sample and population	Blink measurement method	Condition
Barbato et al., 1993, <i>Psychiatry Research</i>	10.1016/0165-1781(93)90057-N	Small clinical subtype study	10 winter SAD patients; control details not fully exposed in snippet	Spontaneous blink-rate measurement, method not exposed in snippet	Before week r photot
Kojima et al., 2002, <i>Psychiatry Clin Neurosci</i>	10.1046/j.1440-1819.2002.01052.x	Pilot case-control	11 male panic-disorder patients + 16 age-matched controls	Blink rate examined during rest and video scenes; exact sensor not exposed in accessible abstract	Rest + audiov stimuli
Rubin et al., 2017, <i>Brain Sciences</i>	10.3390/brainsci7020016	Case-control psychophysiology	68 women in PTSD, trauma-exposed-no-PTSD, and control groups	Blink was digitally recorded by the experimenter using a counter during task performance	Flanked inhibitory control under manip

Source	DOI	Study type	Sample and population	Blink measurement method	Condition
Lee et al., 2024, <i>BMC Geriatrics</i>	10.1186/s12877-024-05034-w	Cross-sectional case-control	25 LLD, 29 healthy older, 25 healthy younger adults	EyeLink 1000 Plus, binocular, 500 Hz	Dark-room, luminance manipulation to evoke light/dark reflexes
Lee et al., 2025, <i>J Geriatr Psychiatry Neurol</i>	10.1177/08919887251334999	Task-based case-control	NR in accessible abstract	Eye-tracking / pupillometry; exact blink algorithm NR in abstract	Emotion conflict processing task

Source	DOI	Study type	Sample and population	Blink measurement method	Condition
Alghowinem et al., 2013, <i>IEEE ICIP</i>	10.1109/ICIP.2013.6738869	Engineering classification study	Exact participant count NR in accessible abstract	Face-video eye-movement features; GMM + SVM / statistical features + SVM	Structure-preserving non-deep learning intervention scenarios
Al-Gawwam & Benaissa, 2018, <i>IEEE ISSPIT</i>	10.1109/ISSPIT.2018.8642682	Engineering blink-feature classification study	Exact participant count NR in accessible abstract	Webcam-based blink-feature extraction; classifiers included AdaBoost , Naive Bayes, Bagging	Structure-preserving North and Fr tasks

Measurement comparison table

Few papers reported standardized effect sizes for blink outcomes, so the table below lists **reported direction, available performance metrics**, and the most important reliability considerations.

Measurement approach	Representative studies	Temporal resolution / algorithm	What it captured best in this literature	Reported blink effect or performance	Reliability issues and confounds
Manual video coding from interviews	Mackintosh 1983	Video, human coding; sampling rate NR	Broad clinical state change over repeated visits	Depressed patients showed higher blink rate , which normalized with improvement	Scalable only with labor; vulnerable to coder variability; limited temporal microstructure. ⁸
Manual counting during cognitive task	Rubin 2017	Human digital counter during flanker task	Contextual/ task-modulated blink differences in PTSD	PTSD-related increase appeared under low threat , not as a universal group difference	Highly task-sensitive; difficult to automate retrospectively; requires well-controlled elicitation. ⁶
Rest + audiovisual symptom provocation / relaxation scenes	Kojima 2002	Method NR in accessible abstract	Panic-related differences across rest and emotional scenes	Panic group had higher blink rate at rest and during stimulation	Very small sample; all male; scene content likely modulates blink independently of anxiety. ¹⁴
High-resolution binocular eye tracking	Lee 2024	500 Hz EyeLink 1000 Plus	Blink rate plus pupil metrics under carefully controlled luminance	Higher LLD blink rate than older and younger controls; symptom correlations	Better temporal fidelity, but medications, ocular disease, luminance, and age still matter. ¹⁷

Measurement approach	Representative studies	Temporal resolution / algorithm	What it captured best in this literature	Reported blink effect or performance	Reliability issues and confounds
Eye tracking during emotional conflict tasks	Lee 2025	Eye tracking / pupillometry, abstract-level details only	Condition-specific blink modulation during emotional congruency/ incongruency	Blink behavior changed with emotional valence/ incongruency ; accessible summary suggests no simple main group difference	Strong task dependence means portability needs strict paradigm replication. ¹²
Face-video eye-movement features with classical ML	Alghowinem 2013	Video-derived features + GMM/SVM or SVM on statistical descriptors	Binary depression classification from structured recordings	70–75% accuracy	Accuracy reported without clinical ROC operating points; population/ sample details incomplete in accessible summaries. ¹⁵
Blink-feature classification from webcam/ interview tasks	Al-Gawwam & Benaissa 2018	Webcam blink features + AdaBoost / Naive Bayes / Bagging	Task-specific depression screening in structured recordings	AdaBoost about 92% on NorthWind and 88% on FreeForm	Likely dataset/ task overfitting; no exposed sensitivity/ specificity; not validated for clinic-to-clinic generalization. ¹⁶

Measurement approach	Representative studies	Temporal resolution / algorithm	What it captured best in this literature	Reported blink effect or performance	Reliability issues and confounds
Consumer landmark/EAR pipeline	User GitHub doc / OpenWillis-oriented implementation	MediaPipe Face Mesh , EAR, z-scoring, prominence thresholding; typical ~30 fps clips	Practical blink counting from ordinary video	Benchmark in the provided note cites MAE ≈ 4.45 blinks/min on 300 one-minute videos	Useful for prototyping, but lower frame rate than lab trackers and sensitive to pose, lighting, glasses, and occlusion; within-clip normalization hurts portability of absolute thresholds.

Discussion

The central conclusion is straightforward: **blink rate alone is not yet a validated diagnostic biomarker** for PTSD, major depressive disorder, or anxiety disorders. It performs best as a **supporting psychophysiological feature** inside a carefully standardized task or a broader multimodal model. This conclusion follows from three recurring patterns in the evidence: the signal is **context-dependent**, the literature is **heterogeneous across depressive subtypes and tasks**, and clinical studies almost never report the kind of threshold-based operating characteristics that would be required for deployment. ¹⁸

For **depression**, the signal is clearly more promising than for PTSD or anxiety. Elevated blink rate appears in the 1983 video-coded work and in the 2024 late-life depression eye-tracking study, and both are clinically coherent: higher blink associated with current illness burden and improvement over time. But the Ebert study shows why the field has not converged on a simple interpretation: if non-retarded MDD does not differ from controls at baseline, then blink rate may be indexing only certain depressive phenotypes, psychomotor profiles, or treatment-responsive arousal states. The 2025 emotional-conflict paper pushes in the same direction by showing that blink behavior depends on **what the participant is doing**, not just what diagnosis they carry. ¹⁹

For **PTSD**, the evidence is not only sparse but also mechanistically mixed. Rubin et al. provide a meaningful signal, but it emerges under a specific inhibitory-control/threat paradigm and not as a universal cross-context marker. That means a large part of any future PTSD-detection utility would come from **task-evoked modulation**, not from passive baseline monitoring. If a measure only separates groups in a narrow laboratory context, it can still be scientifically valuable, but it is not ready for general-purpose screening.

For **anxiety disorders**, the panic-disorder pilot is informative but not decisive. Elevated blink rate in 11 male patients under rest and audiovisual conditions is a legitimate clue, yet it is too small and too narrow a sample to anchor real detection claims. The literature gap in GAD and social anxiety is especially important because it means we cannot currently tell whether blink-rate effects are **panic-specific, higher-order anxiety-state effects**, or simply a product of small-sample task idiosyncrasies. ¹⁴

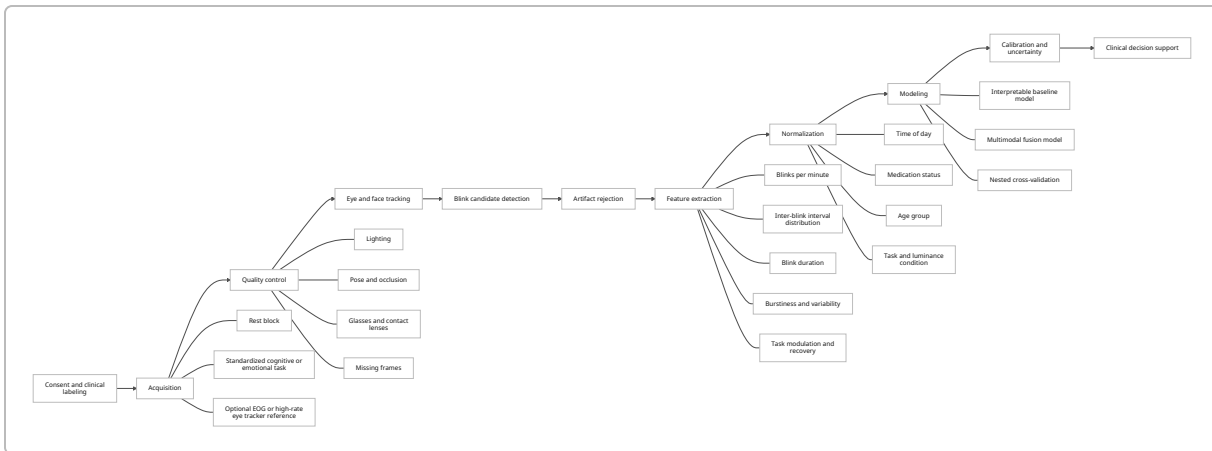
Methodologically, the biggest reliability problem is that blink rate is not a unitary construct. It changes with **task modality, visual load, luminance, time of day, age, medication exposure**, and probably with multiple ocular-surface and fatigue variables. A 2022 psychophysics paper explicitly framed blink patterns as modulated by auditory input, and the 2024 LLD paper itself notes the importance of diurnal variation and medication effects. The user-provided GitHub implementation rightly emphasizes lighting, head pose, glasses, and occlusion as operational constraints. ²⁰

A second conceptual caution is mechanistic interpretation. Several psychiatric papers discuss blink rate as an indirect window into central dopaminergic function, and that framing partly motivates its appeal in depression and panic disorder. But the broader neuroscience literature has questioned how tightly spontaneous eye-blink rate maps onto striatal dopamine in humans; an influential eNeuro paper reported that spontaneous eye-blink rate was **uncorrelated** with dopamine D2 receptor availability. The practical takeaway is that blink rate may still be useful as a marker even if it is **not** a clean dopamine assay. That makes it more appropriate as a **behavioral composite signal** than as a mechanistic surrogate of one neurotransmitter system. ²¹

The engineering literature is encouraging but not dispositive. Accuracies in the 70s and even low 90s show that blink and eye-movement features contain signal under structured conditions. Yet those studies still look much more like **pattern recognition on constrained interview clips** than like real-world psychiatric screening. Without external validation across sites, hardware, age groups, medication strata, and comorbidities, such models should be treated as exploratory. ²²

Practical notes for implementing detection

From an implementation standpoint, blink-based psychiatric detection should be designed as a **multimodal, standardized, quality-controlled workflow**, not as a single-number screening rule. The most evidence-compatible design is to combine **resting blink rate, task-evoked blink modulation**, and neighboring ocular features such as **pupil responses**, then fuse them with non-ocular features such as self-report, speech, or facial movement. This is especially important because the clinical studies that look strongest do not isolate blink rate from task or arousal context. ²³



A practical acquisition protocol should therefore include at least one **resting segment** and one **standardized task segment**. If the goal is PTSD-like detection, a cognitive-control or emotional-conflict paradigm is likely more informative than pure rest, because the strongest PTSD signal in the blink literature is task-conditional. If the goal is depression detection, resting and luminance-controlled segments are useful, but emotional-conflict or inhibitory-control tasks may still add information because blink abnormalities in depression are not purely tonic. ²³

For preprocessing, the provided GitHub note is a reasonable prototype blueprint. It uses **MediaPipe Face Mesh**, calculates **eye aspect ratio**, interpolates missing values, **z-scores EAR within a clip**, and finds blink candidates as prominent troughs with a prominence threshold and minimum frame spacing. The note also recommends **frontal capture**, roughly **50–100 cm** distance, stable ambient light, and warns that glasses, poor illumination, and head motion degrade extraction. Those are sensible engineering choices, but for psychiatric use they need two additions: first, a **hardware- and cohort-specific validation** against manual coding or EOG/eye-tracker reference data; second, preservation of enough raw information to test whether the within-clip normalization itself reduces cross-subject comparability. The note's benchmarked **MAE of about 4.45 blinks/min on 300 one-minute videos** is respectable for consumer video, but that error band may still be large when expected psychiatric differences are modest or context-bound.

Algorithmically, the safest path is to start with **interpretable models**. Use mixed-effects regression or regularized logistic regression on blink-derived features first, stratified by disorder, task, age, and medication band. Only after that should one move to more flexible models. If machine learning is used, the target should not be “blink rate only,” but a richer ocular signature: **rate, duration, inter-blink variability, task modulation, pupil co-variation, and failure-to-detect quality flags**. The depression-classification papers show that blink features can help, but they also imply that the best performance usually comes from **feature sets**, not from a raw blinks-per-minute count. ²⁴

Data requirements should be strict. Training data need clinician-verified labels, repeated sessions, hardware metadata, medication data, and enough sampling across sex, age, and comorbidity to estimate nuisance variance. Most importantly, the model should be trained and evaluated at the **person level**, not just the clip level. Clip-level inflation is a recurring risk in interview/video corpora.

Ethically, eye and face recordings should be treated as **sensitive personal data** because they can simultaneously reveal identity and infer mental-health status. In a clinical setting, blink-derived outputs should be framed as **decision support**, never as an autonomous diagnosis. The practical deployment

standard should therefore be: transparent uncertainty, opt-in consent, secure storage, and a workflow in which a clinician can ignore the model when it conflicts with history, examination, or patient context.

The bottom line for implementation is conservative but constructive: **blink rate is promising as a low-cost auxiliary signal, especially for depression, but it is not sufficiently stable, specific, or standardized to serve alone as a detector of PTSD, MDD, or anxiety disorders.** The highest-value research direction is not “find one blink threshold,” but “build a standardized ocular biomarker panel in which blink dynamics are one component.” ²⁵

¹ ⁴ ⁵ ⁸ ¹⁹ ²⁵ <https://www.cambridge.org/core/journals/the-british-journal-of-psychiatry/article/blink-rate-in-psychiatric-illness/D264FDB3372442BC34918EF8B448D3CC>
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⁶ (no title)

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